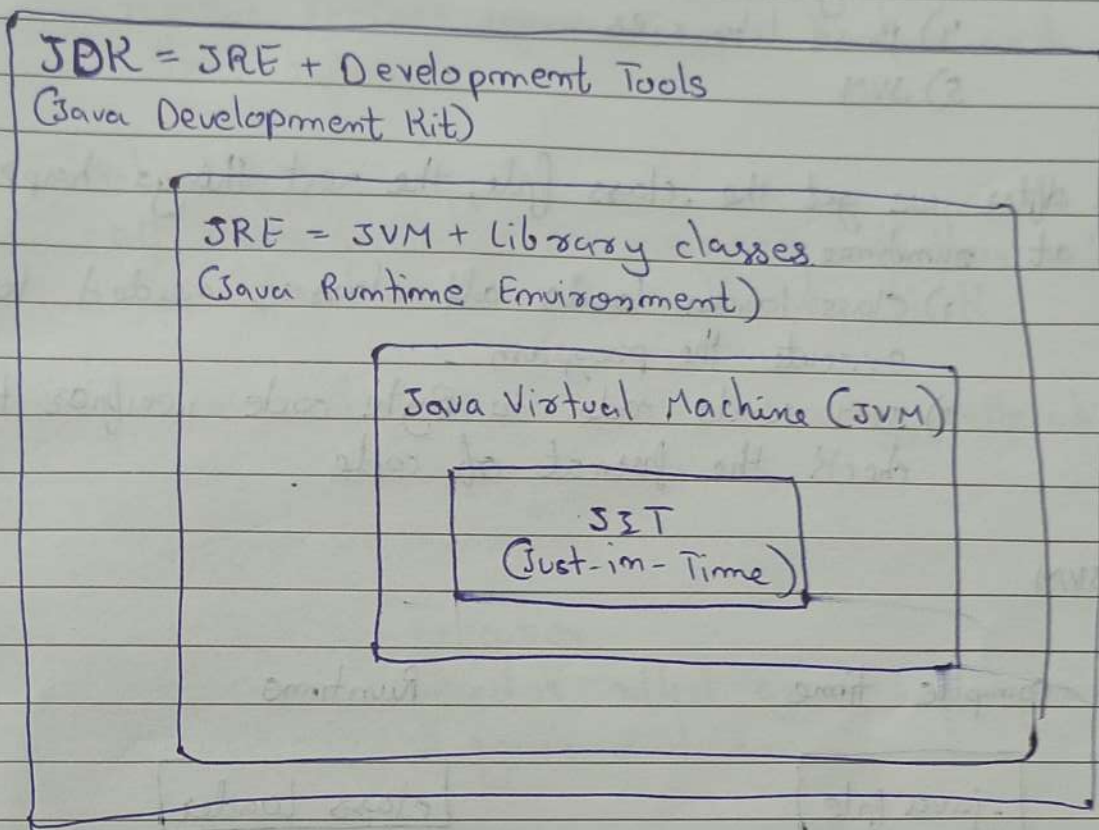


⇒ Architecture



→ JDK

↳ Provides environment to develop and run the Java program.

↳ It is a package that includes:

- 1) development tools - to provide an environment to develop your program.
- 2) JRE - to execute your program
- 3) a compiler - javac
- 4) archiver - jar
- 5) docs generator - javadoc
- 6) interpreter / loader

→ JRE

↳ It is an installation package that provides environment to only run the program.

↳ It consists of:

- 1) Deployment technologies

- 2) User interface toolkits
- 3) Integration libraries
- 4) Base libraries
- 5) JVM

↳ after we get the .class file, the next things happen at runtime:

- 1) Class loader loads all classes needed to execute the program.
- 2) JVM sends code to Byte code verifier to check the format of code.

→ JVM

compile time

.java file

javac
(compilation)

.class file

Runtime

class loader

Byte code verifier

Interpreter

Runtime

Hardware

(How JVM works) class loader.

↳ loading:

- reads .class file and generate binary data
- an object of this class is created in heap.

↳ Linking

- JVM ~~always~~ verifies the .class file.

- allocates memory for class variables & default values
- replace symbolic references from the type with direct reference

↳ Initialization

- all static variables are assigned with their values ~~def~~ defined in the code and static block

↳ JVM contains the stack & Heap memory allocations

JVM Execution

↳ Interpreter

- Line by line execution
- when one method is called again and again, it will interpret again and again

=> JIT

- ↳ those methods that are repeated, JIT provides direct machine code, so re-interpretation is not required.

- ↳ makes execution faster
- ↳ Garbage collector.

